

Application Design and Development (ADaD)

Phase 3: Application Logic –
Command Reference

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**STUDY REAL
BUSINESS.**

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WKO WIEN

JavaScript: Command Reference

1. Using Variables
2. Checking for Conditions
3. Creating a Loop
4. Addressing HTML Elements
5. Manipulating the Text of an HTML Element
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10. Manipulating Numbers

1. Using Variables

```
/* -----  
1. Using variables  
----- */  
  
var variablename;  
console.log(variablename);  
// Outcome: Variable is declared but empty  
  
var anothervariable = 'Initial text';  
console.log(anothervariable);  
// Outcome: Variable is declared and contains now the  
text "Initial text"  
  
variablename = 'Some Text';  
console.log(variablename);  
// Outcome: "Some text" is now stored in the variable  
with the name "variablename"  
  
variablename = 5;  
console.log(variablename);  
// Outcome: Now the variable contains the value 5  
  
variablename = variablename + 5;  
console.log(variablename);  
// Outcome: Now the variable contains the value 10
```

■ Declaring a variable

var variablename;

Outcome: Variable is declared but empty

var anothervariable = "Initial text";

Outcome: Variable is declared and contains now the text "Initial text"

■ Storing some value within a variable

variablename = "Some Text";

Outcome: "Some text" is now stored in the variable with the name "variablename"

variablename = 5;

Outcome: Now the variable contains the value 5

variablename = variablename + 5;

Outcome: Now the variable contains the value 10

2. Checking for Conditions

```
/* -----  
 2. Checking for Conditions  
----- */  
  
var x = 5;  
  
if (x === 5) {  
  console.log('The value of x is 5');  
  // Outputs "The value of x is 5" into the console of  
  // the browser  
} else {  
  console.log('The value of x is not 5');  
  // Does not output anything as the else part is not  
  // triggered  
}
```

■ Syntax

```
if (condition) {  
  doSomething...;  
} else {  
  doSomethingElse...;  
}
```

■ Example

```
var x = 5;  
if (x === 5) {  
  console.log('The value of x is 5'); // Outputs "The  
  value of x is 5" into the console of the browser  
} else {  
  console.log('The value of x is not 5'); // Does not  
  output anything as the else part is not triggered  
}
```

3. Creating a Loop

```
/* -----  
  3. Creating Loops  
----- */  
  
var x = 1;  
  
while (x <= 10) {  
  console.log(x);  
  x = x + 1;  
}  
// Outcome: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
```

- Create a “while”-loop

```
var x = 1;  
while (x <= 10) {  
  console.log(x);  
  x = x + 1;  
}
```

Outcome: 1, 2, 3, 4, 5, 6, 7, 8,
9, 10

4. Addressing HTML Elements



```
/* -----  
  4. Addressing HTML Elements  
----- */  
  
// Selecting an element with an ID  
document.querySelector('#nameoftheid');  
  
// Selecting an element with a CLASS  
document.querySelector('.nameoftheclass');  
  
// Selecting multiple elements with the same CLASS  
document.querySelectorAll('.nameoftheclass');
```

- Selecting an element with an ID:
document.querySelector('#nameoftheid');
;
- Selecting an element with a CLASS:
document.querySelector('.nameoftheclass');
- Selecting multiple elements with the same CLASS:
document.querySelectorAll('.nameoftheclass');

5. Manipulating the Text of an HTML Element

```
/* -----  
   5. Manipulating the Text of an HTML Element  
   ----- */  
  
// Select the element (see 4.)  
var element = document.querySelector('#nameoftheid');  
  
// Fetch the text  
var text = element.textContent;  
  
// Overwrite the text  
element.textContent = 'Some other text';
```

- Select the element (see 4.)
*var element =
document.querySelector('#name
eoftheid');*
- Fetch the text
var text = element.textContent;
- Overwrite the text
*element.textContent = 'Some
other text';*

6. Manipulating Value of an HTML INPUT Element

```
/* -----  
   6. Manipulating the Value of an HTML INPUT Element  
   ----- */  
  
// Select the element (see 4.)  
var element = document.querySelector('#idoftheinput');  
  
// Fetch the value  
var value = element.value;  
  
// Overwrite the text  
element.value = 'Some other value';
```

- Select the element (see 4.)
*var element =
document.querySelector('#idoftheinput');*
- Fetch the value
var value = element.value;
- Overwrite the text
element.value = 'Some other value';

7. Creating HTML Elements in JavaScript

```
/* -----  
 7. Creating HTML Elements  
----- */  
  
// Create a list element  
// Example: <li>Text</li>  
var li = document.createElement('li');  
li.textContent = 'Text';  
console.log(li);  
  
// Create a link  
// Example: <a href="www.xyz.com">Linktext</a>  
var link = document.createElement('a');  
link.href = 'www.xyz.com';  
link.textContent = 'Linktext';  
console.log(link);
```

- Create a list element

Example: `<li>Text</li>`

```
var li = document.createElement('li');
```

```
li.textContent = 'Text';
```

- Create a link

Example: `<a`

`href="www.xyz.com">Linktext</a>`

```
var link = document.createElement('a');
```

```
link.href = 'www.xyz.com';
```

```
link.textContent = 'Linktext';
```

8. Adding HTML Elements to a HTML Document in JavaScript

```
/* -----  
 8. Adding HTML Elements  
----- */  
  
// Create a list element  
// Example: <li>Text</li>  
var li = document.createElement('li');  
li.textContent = 'Text';  
  
// Add the element to the HTML document  
var list = document.querySelector('#listid');  
list.appendChild(li);
```

- Create a list element (see 7.)
Example: `<li>Text</li>`
var li =
document.createElement('li');
li.textContent = 'Text';
- Add the element to the HTML document
var list =
document.querySelector('#listid');
list.appendChild(li);

9. Reacting to DOM-Events in JavaScript



```
/* -----  
  9. Reacting to DOM-Events  
----- */  
  
// Select the element (see 4.)  
var element = document.querySelector('#nameoftheid');  
  
// Bind a "click-listener" to the element  
element.addEventListener('click', function () {  
    console.log('Hi');  
});
```

- Select the element (see 4.)
*var element =
document.querySelector('#name
eoftheid');*
- Bind a “click-listener” to the element
*element.addEventListener('click'
, function () {
doSomething...;
});*

10. Manipulating Numbers

- Get the value of an input (see 6.)

```
var element = document.querySelector('#nameoftheid');  
var value = element.value;
```

- Interpret the value as an Integer

```
value = parseInt(value, 10);
```

- Check if a value is not a Number

```
if (!Number.isNaN(value)) {  
    console.log('Value is not a number');  
}
```